

Apex Heavy Engineering Pvt Ltd

ENGINEERING TECHNICAL DATASHEET & COMPLIANCE MATRIX • TENDER REF: CPCL/ENG/2026/HPGC-0412

API 618 5TH EDITION RECIPROCATING GAS COMPRESSOR DATASHEET

Design Specification: Manali Refinery High-Pressure Gas Compression System

Clause 5.4

Engineering Parameter	Tender Specified Mandatory Value	Bidder Offered Value	Compliance Status
Compressor Type & Configuration	API 618 Balanced-Opposed Reciprocating	API 618 5th Ed 2-Stage Balanced Opposed	COMPLIANT
Operating Discharge Pressure	Minimum 120.0 bar(g) continuous	120.0 bar(g) rated (140.0 bar(g) MAWP)	COMPLIANT
Suction Operating Pressure	18.5 bar(g) nominal refinery gas feed	18.5 bar(g) sweet natural gas mix	COMPLIANT
Driver Motor Rating & Classification	Minimum 1,800 kW, Flameproof Ex d IIB+H2	1,850 kW, 6.6 kV, Ex d IIB+H2 T4	COMPLIANT
Dry Gas Seal Barrier Arrangement	Dual pressurized nitrogen buffer seal	Dual pressurized nitrogen barrier (API 682)	COMPLIANT
Vibration Monitoring System	Eddy current proximity probes per API 670	API 670 Bently Nevada Proximitor Probes	COMPLIANT

SAFETY INTEGRITY (SIL-3) & AUXILIARY SYSTEM ENGINEERING SCHEDULE

Emergency Shutdown System (ESD), Dry Gas Seals & API 614 Lubrication

Clause 5.4 Schedule B

Subsystem / Module	Engineering Design Standard	Offered Specification	Safety Compliance
Emergency Shutdown System (ESD)	IEC 61508 / 61511 Standards	SIL-3 Certified 2oo3 Triple Modular Redundant PLC	FULL CONFORMANCE
Forced-feed Lube Oil System	API 614 Standard Package	Dual 100% pumps, duplex filters, SS coolers	FULL CONFORMANCE
Pulsation Suppression Dampeners	API 618 Design Approach 3	Suction & discharge pulsation bottles per DA-3	FULL CONFORMANCE
Skid Structural Engineering	Rigid structural baseplate	Heavy duty structural steel with drip lip	FULL CONFORMANCE

CHIEF TECHNICAL OFFICER VERIFICATION & ENGINEERING STAMP

Name: P. Ramaswamy
Designation: Chief Engineer (Turbomachinery & Process Systems)
Entity: Apex Heavy Engineering Pvt Ltd
Date & Time: 04-Sep-2026 • Verified Engineering Drawings Attached
Ref / UDIN / Reg: TECH-API618-CONF-0412

